

7.2.3 Software Quality Assurance Process

NOTE — The Software Quality Assurance (SQA) Process provides assurance that work products and processes comply with predefined provisions, standards, and plans. It is closely coordinated with supporting processes such as Software Verification, Software Validation, Software Review, Software Audit, and Software Problem Resolution.

7.2.3.1 Purpose

The purpose of the Software Quality Assurance Process is to **provide assurance that work products and processes comply with predefined provisions and plans**, ensuring conformance to contractual, regulatory, and organizational requirements.

7.2.3.2 Outcomes

As a result of successful implementation of the Software Quality Assurance Process:

- a) a strategy for conducting quality assurance is developed;
 - b) evidence of software quality assurance is produced and maintained;
 - c) problems and/or non-conformances with requirements are identified and recorded; and
 - d) adherence of products, processes, and activities to the applicable standards, procedures, and requirements is verified.
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7.2.3.3 Activities and Tasks

The project shall implement the following activities in accordance with applicable organizational policies and procedures with respect to the Software Quality Assurance Process.

7.2.3.3.1 Process Implementation

This activity consists of the following tasks:

7.2.3.3.1.1

A quality assurance process suited to the project shall be established. The objectives shall be to ensure that software products and processes comply with their established requirements and adhere to their established plans.

7.2.3.3.1.2

The quality assurance process shall be coordinated with the related **Software Verification (7.2.4)**, **Software Validation (7.2.5)**, **Software Review (7.2.6)**, and **Software Audit (7.2.7)** processes to ensure comprehensive quality oversight.

7.2.3.3.1.3

A **Software Quality Assurance Plan** shall be developed, documented, implemented, and maintained for the life of the project or contract. The plan shall include:

- a) quality standards, methodologies, procedures, and tools for performing the quality assurance activities (or references to official organizational documentation);
- b) procedures for contract review and coordination thereof;
- c) procedures for identification, collection, filing, maintenance, and disposition of quality records;
- d) resources, schedule, and responsibilities for conducting quality assurance activities;
- e) selected activities and tasks from supporting processes such as Software Verification (7.2.4), Software Validation (7.2.5), Software Review (7.2.6), Software Audit (7.2.7), and Software Problem Resolution (7.2.8).

7.2.3.3.1.4

Scheduled and on-going quality assurance activities and tasks shall be executed. When problems or non-conformances with contract or project requirements are detected, they shall be documented and serve as input to the **Software Problem Resolution Process (7.2.8)**. Records of these activities, their execution, problems, and resolutions shall be prepared and maintained.

7.2.3.3.1.5

Records of quality assurance activities and tasks shall be made available to the acquirer or designated authority, as specified in the contract.

7.2.3.3.1.6

It shall be assured that persons responsible for assuring compliance with the contract requirements have the organizational freedom, resources, and authority to perform objective evaluations and to initiate, effect, resolve, and verify problem resolutions.

7.2.3.3.2 Product Assurance

This activity consists of the following tasks:

7.2.3.3.2.1

It shall be assured that all plans required by the contract are documented, comply with the contract, are mutually consistent, and are being executed as required.

7.2.3.3.2.2

It shall be assured that software products and related documentation comply with the contract and adhere to the approved plans.

7.2.3.3.2.3

In preparation for delivery of the software products, it shall be assured that they have fully satisfied their contractual requirements and are acceptable to the acquirer.

7.2.3.3.3 Process Assurance

This activity consists of the following tasks:

7.2.3.3.3.1

It shall be assured that internal software engineering practices, development environments, test environments, and libraries comply with contract requirements, applicable standards, and organizational procedures.

7.2.3.3.3.2

It shall be assured that applicable prime-contract requirements are flowed down to subcontractors, and that subcontractors' software products satisfy prime-contract requirements.

7.2.3.3.3.3

It shall be assured that the acquirer and other parties are provided with the required support and cooperation in accordance with the contract, negotiations, and project plans.

7.2.3.3.3.4

It shall be assured that software product and process measurements are performed in accordance with established standards and procedures.

7.2.3.3.5

It shall be assured that staff assigned to the project possess the necessary skills and knowledge to meet the project requirements and receive required training as needed.

7.2.3.3.4 Assurance of Quality Systems

This activity consists of the following task:

7.2.3.3.4.1

Additional quality management activities may be performed in accordance with the clauses of **ISO 9001** or equivalent organizational quality management standards to ensure alignment between the project's quality assurance activities and broader organizational quality objectives.

7.2.3.4 Deliverables

The following deliverables shall be produced during this process:

- Software Quality Assurance Plan;
 - Records of QA activities and audits;
 - Non-conformance reports and problem tracking logs;
 - Quality review reports and metrics;
 - Evidence of coordination with verification, validation, review, and audit processes;
 - Qualification and training records for QA personnel.
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7.2.3.5 Audit and Configuration Control

All SQA artifacts (plans, records, reports, logs) shall be placed under appropriate configuration management. Audit evidence shall be retained for at least one release cycle or in accordance with organizational and regulatory requirements. QA records shall be protected against unauthorized alteration or deletion, and access shall be controlled and logged.

Cross-References

This process interacts with the following:

- **Software Requirements Analysis Process (7.1.2)** — for traceability and input requirements.
- **Software Construction Process (7.1.5)** — to verify adherence to coding standards and construction plans.
- **Software Integration Process (7.1.6)** — to assure integration procedures and outcomes comply with requirements.
- **Software Qualification Testing Process (7.1.7)** — to verify testing results and ensure quality evidence.
- **Software Verification Process (7.2.4)** — for coordination of test evidence and evaluation activities.
- **Software Validation Process (7.2.5)** — to ensure QA evidence aligns with validation outcomes.
- **Software Review (7.2.6)** and **Software Audit (7.2.7)** — for formal inspections and audits.
- **Software Problem Resolution Process (7.2.8)** — for handling and tracking non-conformances.